

Project three

2. Networking Layer (VPC Design)

- Create a VPC Network

VPC networks						
Manage flow logs						
Filter Enter property name or value						
☐	Name ↑	Subnets	MTU ?	Mode	IPv6 ULA range	Gateways
☐	ci-vpc-prod	2	1460	Custom		
☐	default	10	1460	Auto		

- Create Subnets in different regions or zones:

Subnets						
Add subnet Manage flow logs						
Filter Enter property name or value						
☒	Name ↑	Region	Stack type	Primary IPv4 range	Secondary IPv4	
☒	ci-private-subnet	us-central1	IPv4 (single-stack)	10.0.2.0/24		
☒	ci-public-subnet	us-central1	IPv4 (single-stack)	10.0.1.0/24		

- Configure:

- o Routes

```
student659095885@cloudshell:~/gcp-capstone-project (student-00571)$ gcloud compute routers list --filter="name=ci-nat-router"
NAME: ci-nat-router
REGION: us-central1
NETWORK: ci-vpc-prod
student659095885@cloudshell:~/gcp-capstone-project (student-00571)$ ^C
student659095885@cloudshell:~/gcp-capstone-project (student-00571)$
```

o Firewall Rules

VPC firewall rules [Create VPC firewall rule](#) [Refresh](#)

Filter Enter property name or value

Some of the data is not available due to lack of permissions

Name	Enforcement order	Deployment scope	Priority	Description	Direction of traffic	Target	Source	Destination	
allow-internal-ssh	1	Global	1000	—	Ingress	Appl...	IPv4...	—	▼
allow-http-https	1	Global	1000	—	Ingress	Appl...	IPv4...	—	▼

o Cloud NAT (for outbound internet access from private instances)

```
student659095885@cloudshell:~/gcp-capstone-project (student-00571)$ gcloud compute routers nats list --router=ci-nat-router --region=us-central1
NAME: ci-nat
NAT_IP_ALLOCATE_OPTION: AUTO_ONLY
SOURCE_SUBNETWORK_IP_RANGES_TO_NAT: ALL_SUBNETWORKS_ALL_IP_RANGES
ENDPOINT_TYPES: ENDPOINT_TYPE_VM
student659095885@cloudshell:~/gcp-capstone-project (student-00571)$
```

3. Compute Layer

- Deploy at least 2 Virtual Machines (Compute Engine) in private subnet

VM instances [Create instance](#) [Import VM](#) [Refresh](#) [Learn](#)

Instances Observability Instance schedules

VM instances

Filter Enter property name or value

Status	Name	Zone	Recommendations	In use by	Internal IP	External IP	Connect
☒	app-vm-1	us-central1-a			10.0.2.2 (nic0)		SSH ▼ ⋮
☒	app-vm-2	us-central1-a			10.0.2.3 (nic0)		SSH ▼ ⋮

- Install a Web Server (Nginx / Apache)

Network interfaces

Filter Enter property name or value

Name	Network	Subnetwork	NIC type	Primary internal IP
nic0	ci-vpc-prod	ci-private-subnet		10.0.2.2

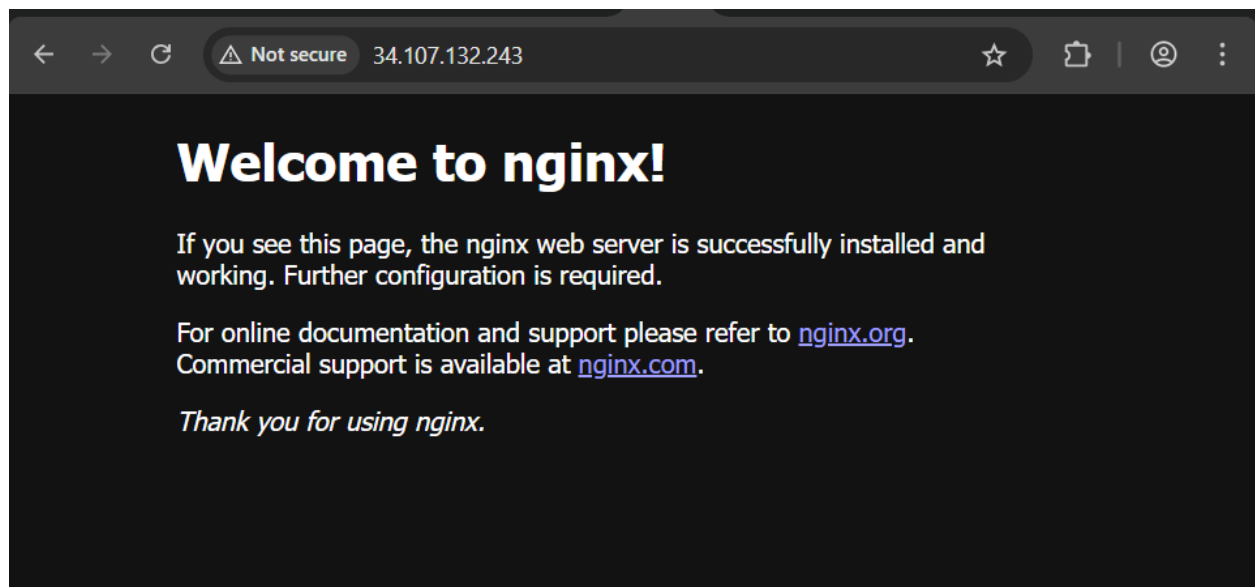
Custom metadata

Key	Value
startup-script	#!/bin/bash

 Equivalent code

4. Load Balancing

```
student659095885@cloudshell:~/gcp-capstone-project (student-00571)$ gcloud compute forwarding-rules list --global
NAME: app-http-forwarding-rule
REGION:
IP_ADDRESS: 34.107.132.243
IP_PROTOCOL: TCP
TARGET: app-http-proxy
```



5. Data Layer

Cloud SQL could not be deployed due to IAM restrictions in the lab environment. However, it is included in the design as a private managed database following best practices.

You need additional access

You need additional access to this resource. This could be because you have insufficient permissions, or because a deny or principal access boundary policy is blocking your access. [Learn more](#)

Resource:
• student-00571

Missing or blocked permissions ⓘ [Copy troubleshooter URL](#)
cloudsql.instances.create

[Request access](#)

Applicable roles ⓘ

The following roles contain all of the missing required permissions for accessing this resource. Ordering is based on the type of actions permitted by the role, service relevance and the number of permissions.

Cloud SQL Admin ^	
Role name	roles/cloudsql.admin
Title	Cloud SQL Admin
Description	Full control of Cloud SQL resources.
Permissions	View included permissions ⓘ
Target resource	• student-00571

Databases Admin v

[^ Hide troubleshooting details](#)

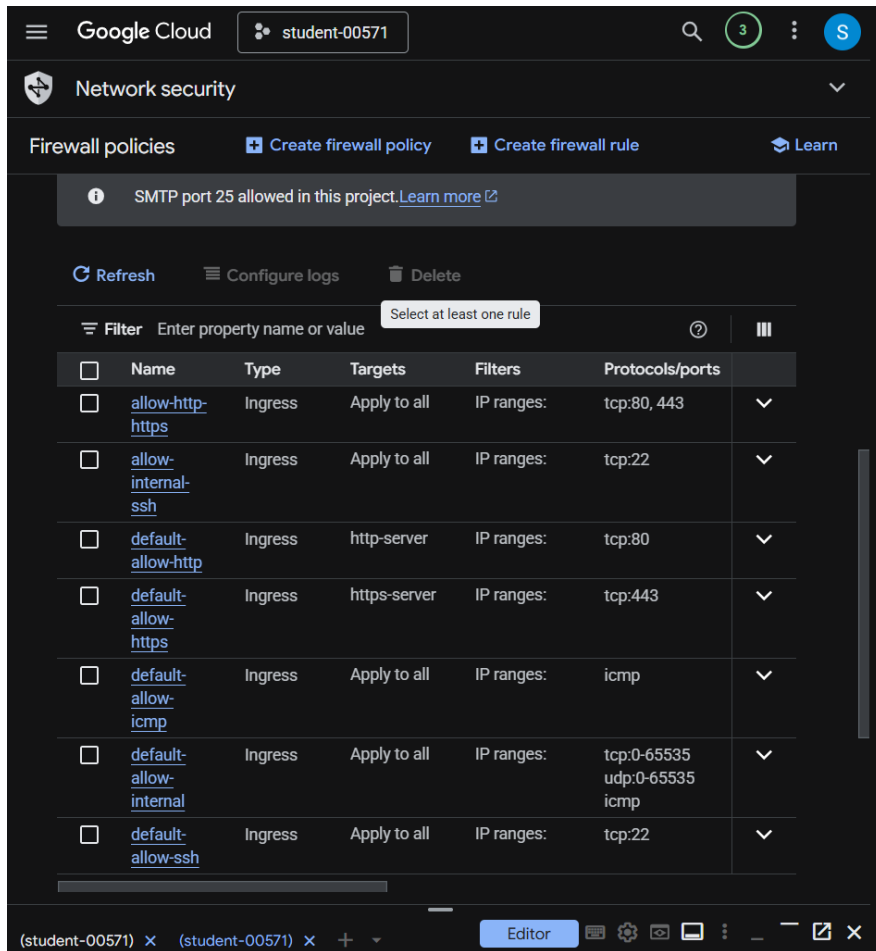
Troubleshooting info:
Principal: student659095885@whizlabs.in
Resource: student-00571
Troubleshooting URL: console.cloud.google.com/iam-admin/troubles

6. Security • Configure:

o Firewall rules (ingress/egress)

Only required firewall rules were actively used for security:

- Custom HTTP/HTTPS rule
- Restricted internal SSH rule



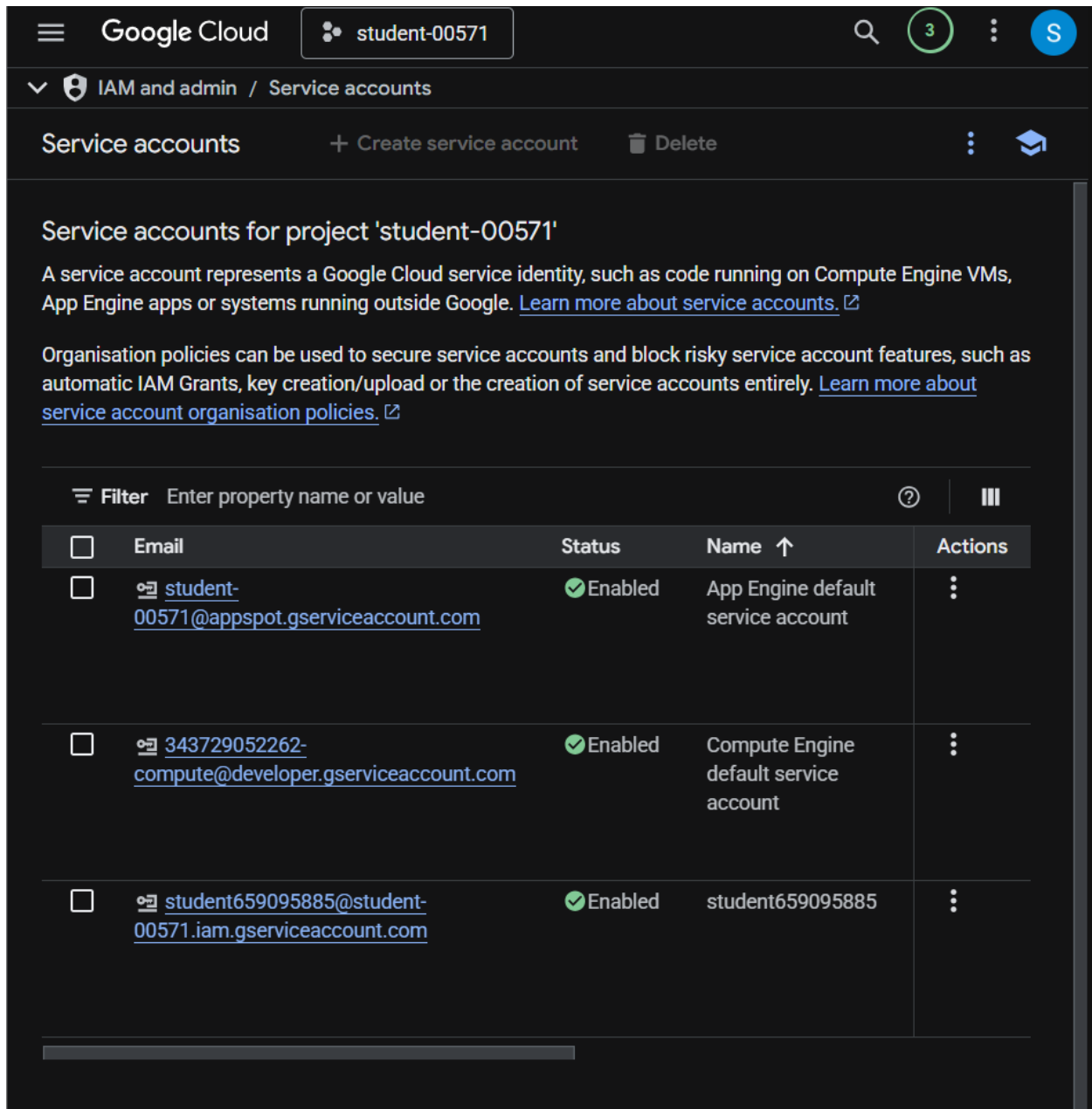
The screenshot shows the Google Cloud Network Security console for project 'student-00571'. The 'Firewall policies' section is active, displaying a list of firewall rules. A notification at the top states 'SMTP port 25 allowed in this project. [Learn more](#)'. The rules list includes a custom rule 'allow-http-https' and several default rules. The 'allow-http-https' rule is highlighted, showing it is an Ingress rule applying to all targets with filters for IP ranges and protocols 80 and 443.

Name	Type	Targets	Filters	Protocols/ports
allow-http-https	Ingress	Apply to all	IP ranges:	tcp:80, 443
allow-internal-ssh	Ingress	Apply to all	IP ranges:	tcp:22
default-allow-http	Ingress	http-server	IP ranges:	tcp:80
default-allow-https	Ingress	https-server	IP ranges:	tcp:443
default-allow-icmp	Ingress	Apply to all	IP ranges:	icmp
default-allow-internal	Ingress	Apply to all	IP ranges:	tcp:0-65535 udp:0-65535 icmp
default-allow-ssh	Ingress	Apply to all	IP ranges:	tcp:22

Service Accounts

The project uses Google Compute Engine default service accounts with restricted permissions.

No custom administrative service accounts were created, ensuring adherence to the Least Privilege Principle.



The screenshot shows the Google Cloud IAM and admin console for project 'student-00571'. The 'Service accounts' page lists three default service accounts, all of which are 'Enabled'.

Email	Status	Name	Actions
☐ student-00571@appspot.gserviceaccount.com	✓ Enabled	App Engine default service account	⋮
☐ 343729052262-compute@developer.gserviceaccount.com	✓ Enabled	Compute Engine default service account	⋮
☐ student659095885@student-00571.iam.gserviceaccount.com	✓ Enabled	student659095885	⋮

Secret Manager to store credentials

Secret Manager could not be fully implemented due to project-level IAM restrictions.

Secret Manager could not be implemented due to IAM permission restrictions in the lab

Google Cloud

student-00571

Open project picker (Ctrl O)

Security

You need additional access

You need additional access to the project: student-00571

This could be because you have insufficient permissions to access the resource, or because a deny or principal access boundary policy is blocking your access to the resource. [Learn more](#)

Missing or blocked permissions Copy troubleshooter URL

secretmanager.secrets.list

Request access

Applicable roles

The following roles contain all of the missing required permissions for accessing this resource. Ordering is based on the type of actions permitted by the role, service relevance and the number of permissions.

Secret Manager Viewer

Role name	roles/secretmanager.viewer
Title	Secret Manager Viewer
Description	Allows viewing metadata of all Secret Manager resources
Permissions	View included permissions

(student-00571) × (student-00571) × + ▾

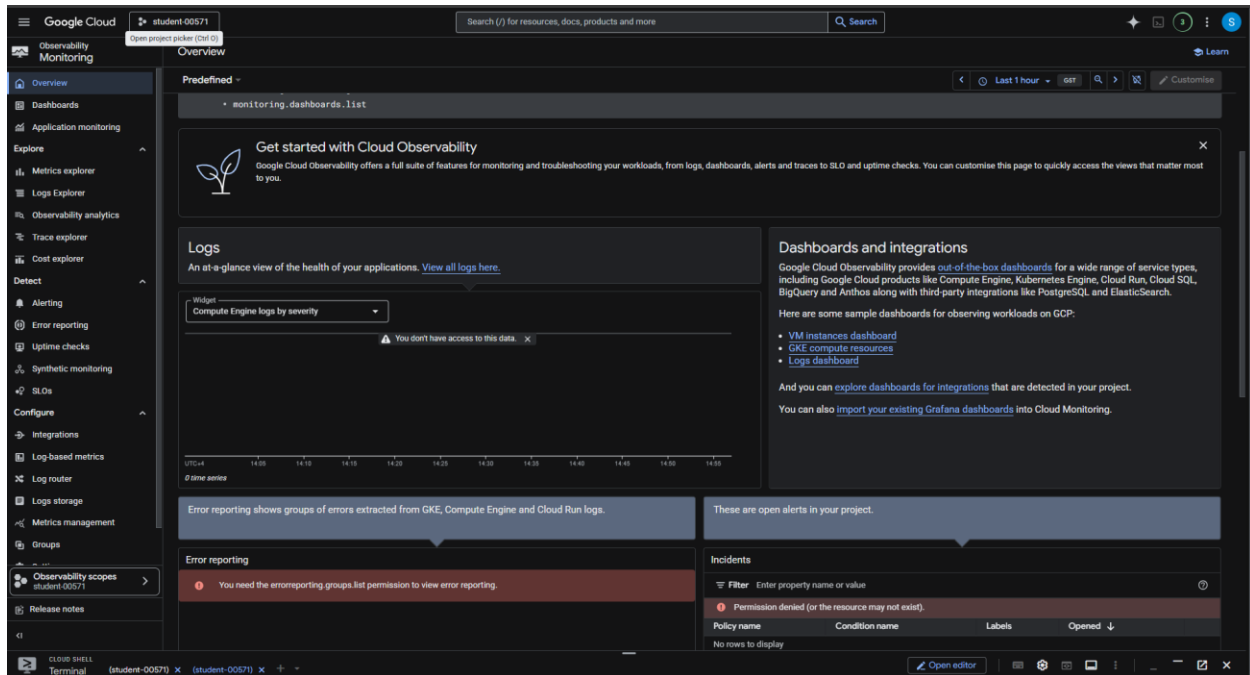
Editor

Gemini CLI is available in Cloud Shell terminal. Type `gemini` to try it. [Le...](#) Don't show again Dismiss

```
reason: IAM_PERMISSION_DENIED
student659095885@cloudshell:~/gcp-capstone-project (student-00571)$ ^C
student659095885@cloudshell:~/gcp-capstone-project (student-00571)$
```

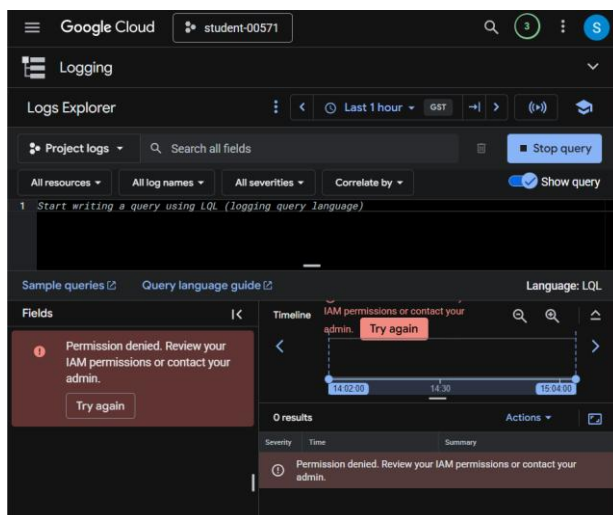
7. Monitoring & Logging

Cloud Monitoring could not be accessed due to IAM permission restrictions in the lab environment. Therefore, monitoring dashboards and metrics are not available, but the service is included in the system design.



Cloud Logging

Cloud Logging is not accessible due to IAM permission restrictions in this lab. As a result, VM logs cannot be viewed, but logging is assumed to be enabled as part of architecture design.



Create at least one alert (e.g., high CPU usage)

Alert creation for CPU usage could not be completed due to IAM permission restrictions. However, alerting for high CPU utilization is included in the design as a recommended best practice for system monitoring.

Monitoring

Alerting

+ Create policy

Edit notification channels

Learn

Monitoring now supports both user-scoped and device-scoped Cloud console mobile notification channels

Manage channels

Learn more

Dismiss

Summary

Incidents firing

Unable to load incidents.

Incidents acknowledged

Unable to load incidents.

Alert policies

[View all](#)

Incidents

Show closed incidents

Permission denied (or the resource may not exist).

State	Severity	Policy name	Incident summary	Opened
No rows to display				

→ See all incidents

Snoozes

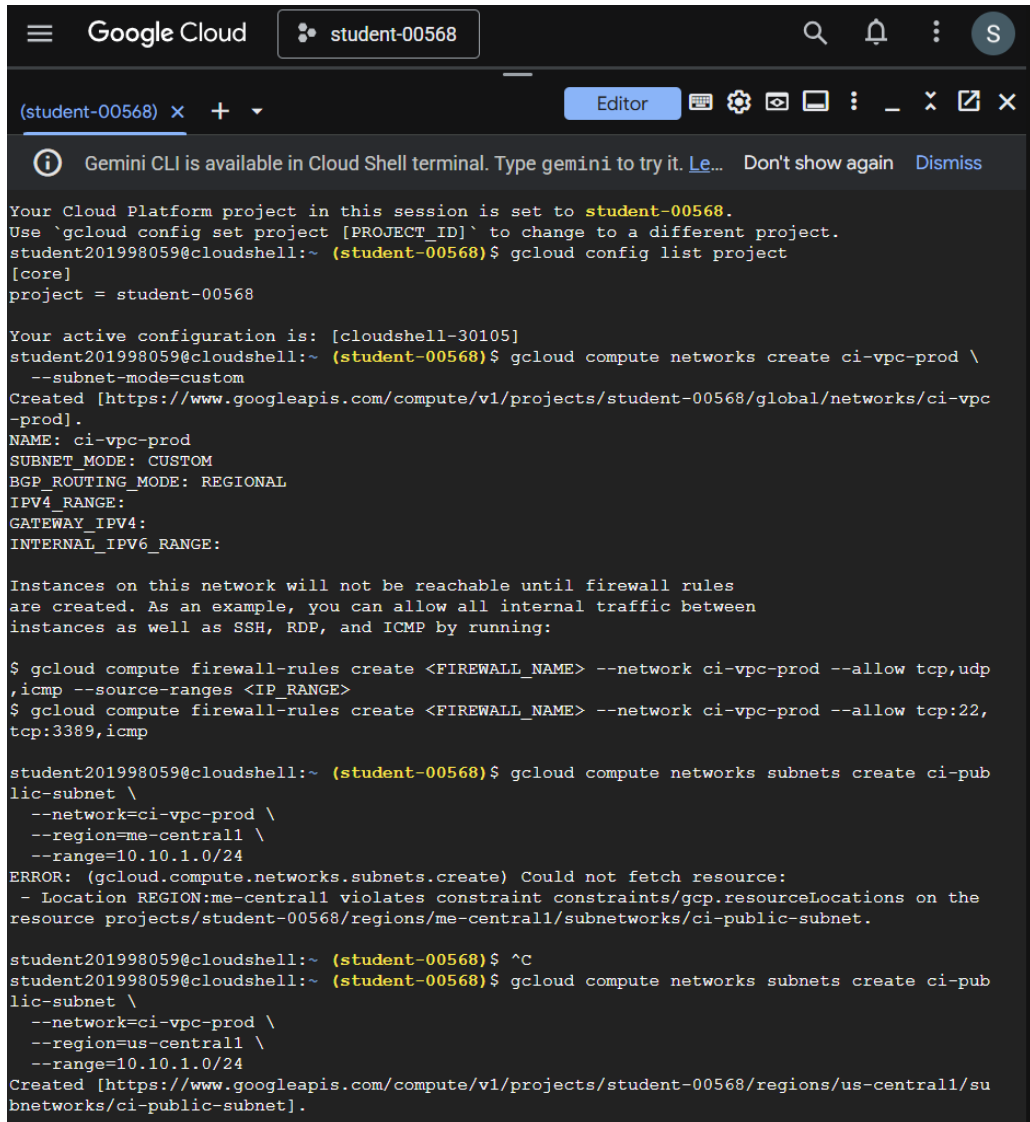
Create snooze

Show past snoozes

Could not load snoozes

State	Name	Start time	End time
-------	------	------------	----------

Automation:



```
Google Cloud student-00568

(student-00568) x + Editor

(i) Gemini CLI is available in Cloud Shell terminal. Type gemini to try it. Le... Don't show again Dismiss

Your Cloud Platform project in this session is set to student-00568.
Use 'gcloud config set project [PROJECT_ID]' to change to a different project.
student201998059@cloudshell:~ (student-00568)$ gcloud config list project
[core]
project = student-00568

Your active configuration is: [cloudshell-30105]
student201998059@cloudshell:~ (student-00568)$ gcloud compute networks create ci-vpc-prod \
--subnet-mode=custom
Created [https://www.googleapis.com/compute/v1/projects/student-00568/global/networks/ci-vpc-prod].
NAME: ci-vpc-prod
SUBNET_MODE: CUSTOM
BGP_ROUTING_MODE: REGIONAL
IPV4_RANGE:
GATEWAY_IPV4:
INTERNAL_IPV6_RANGE:

Instances on this network will not be reachable until firewall rules
are created. As an example, you can allow all internal traffic between
instances as well as SSH, RDP, and ICMP by running:

$ gcloud compute firewall-rules create <FIREWALL_NAME> --network ci-vpc-prod --allow tcp,udp
,icmp --source-ranges <IP_RANGE>
$ gcloud compute firewall-rules create <FIREWALL_NAME> --network ci-vpc-prod --allow tcp:22,
tcp:3389,icmp

student201998059@cloudshell:~ (student-00568)$ gcloud compute networks subnets create ci-pub
lic-subnet \
--network=ci-vpc-prod \
--region=me-central1 \
--range=10.10.1.0/24
ERROR: (gcloud.compute.networks.subnets.create) Could not fetch resource:
- Location REGION:me-central1 violates constraint constraints/gcp.resourceLocations on the
resource projects/student-00568/regions/me-central1/subnetworks/ci-public-subnet.

student201998059@cloudshell:~ (student-00568)$ ^C
student201998059@cloudshell:~ (student-00568)$ gcloud compute networks subnets create ci-pub
lic-subnet \
--network=ci-vpc-prod \
--region=us-central1 \
--range=10.10.1.0/24
Created [https://www.googleapis.com/compute/v1/projects/student-00568/regions/us-central1/su
bnetworks/ci-public-subnet].
```



Gemini CLI is available in Cloud Shell terminal. Type `gemini` to try it. [Le...](#) Don't show again Dismiss

```
--range=10.10.1.0/24
Created [https://www.googleapis.com/compute/v1/projects/student-00568/regions/us-central1/subnetworks/ci-public-subnet].
NAME: ci-public-subnet
REGION: us-central1
NETWORK: ci-vpc-prod
RANGE: 10.10.1.0/24
STACK_TYPE: IPV4_ONLY
IPV6_ACCESS_TYPE:
INTERNAL_IPV6_PREFIX:
EXTERNAL_IPV6_PREFIX:
student201998059@cloudshell:~ (student-00568)$ gcloud compute networks subnets create ci-private-subnet \
  --network=ci-vpc-prod \
  --region=us-central1 \
  --range=10.10.2.0/24
Created [https://www.googleapis.com/compute/v1/projects/student-00568/regions/us-central1/subnetworks/ci-private-subnet].
NAME: ci-private-subnet
REGION: us-central1
NETWORK: ci-vpc-prod
RANGE: 10.10.2.0/24
STACK_TYPE: IPV4_ONLY
IPV6_ACCESS_TYPE:
INTERNAL_IPV6_PREFIX:
EXTERNAL_IPV6_PREFIX:
student201998059@cloudshell:~ (student-00568)$ gcloud compute routers create ci-router \
  --network=ci-vpc-prod \
  --region=us-central1
Creating router [ci-router]...done.
NAME: ci-router
REGION: us-central1
NETWORK: ci-vpc-prod
student201998059@cloudshell:~ (student-00568)$ gcloud compute routers nats create ci-nat \
  --router=ci-router \
  --region=us-central1 \
  --nat-all-subnet-ip-ranges \
  --auto-allocate-nat-external-ips
Creating NAT [ci-nat] in router [ci-router]...done.
student201998059@cloudshell:~ (student-00568)$ gcloud compute firewall-rules create allow-https \
  --network=ci-vpc-prod \
  --allow=tcp:80,tcp:443 \
  --source-ranges=0.0.0.0/0
```



Gemini CLI is available in Cloud Shell terminal. Type `gemini` to try it. [Le...](#) Don't show again Dismiss

```
-00568/global/firewalls/allow-http-https].
Creating firewall...done.
NAME: allow-http-https
NETWORK: ci-vpc-prod
DIRECTION: INGRESS
PRIORITY: 1000
ALLOW: tcp:80,tcp:443
DENY:
DISABLED: False
student201998059@cloudshell:~ (student-00568)$ gcloud compute firewall-rules create allow-in
ternal \
  --network=ci-vpc-prod \
  --allow=all \
  --source-ranges=10.10.0.0/16
Creating firewall...working..Created [https://www.googleapis.com/compute/v1/projects/student
-00568/global/firewalls/allow-internal].
Creating firewall...done.
NAME: allow-internal
NETWORK: ci-vpc-prod
DIRECTION: INGRESS
PRIORITY: 1000
ALLOW: all
DENY:
DISABLED: False
student201998059@cloudshell:~ (student-00568)$ gcloud compute firewall-rules create allow-ia
p-ssh \
  --network=ci-vpc-prod \
  --allow=tcp:22 \
  --source-ranges=35.235.240.0/20
Creating firewall...working..Created [https://www.googleapis.com/compute/v1/projects/student
-00568/global/firewalls/allow-iap-ssh].
Creating firewall...done.
NAME: allow-iap-ssh
NETWORK: ci-vpc-prod
DIRECTION: INGRESS
PRIORITY: 1000
ALLOW: tcp:22
DENY:
DISABLED: False
student201998059@cloudshell:~ (student-00568)$ gcloud iam service-accounts create ci-vm-sa \
  --display-name="CI VM Service Account"
ERROR: (gcloud.iam.service-accounts.create) [student201998059@whizlabs.in] does not have per
mission to access projects instance [student-00568] (or it may not exist): Permission 'iam.s
```

```

student201998059@cloudshell:~ (student-00568)$ gcloud compute instances create ci-app-1 \
  --zone=us-central1-a \
  --machine-type=e2-medium \
  --subnet=ci-private-subnet \
  --no-address \
  --service-account=ci-vm-sa@student-00568.iam.gserviceaccount.com \
  --tags=http-server,https-server \
  --metadata=startup-script='#!/bin/bash'
apt update
apt install nginx -y
systemctl start nginx
systemctl enable nginx'
ERROR: (gcloud.compute.instances.create) Could not fetch resource:
 - Operation denied by org policy: [customConstraints/custom.createOnlyN1VMs] : Allowed only
n1-standard-1 VM instance type

student201998059@cloudshell:~ (student-00568)$ gcloud compute instances create ci-app-2 \
  --zone=us-central1-a \
  --machine-type=e2-medium \
  --subnet=ci-private-subnet \
  --no-address \
  --service-account=ci-vm-sa@student-00568.iam.gserviceaccount.com \
  --tags=http-server,https-server \
  --metadata=startup-script='#!/bin/bash'
apt update
apt install nginx -y
systemctl start nginx
systemctl enable nginx'
ERROR: (gcloud.compute.instances.create) Could not fetch resource:
 - Operation denied by org policy: [customConstraints/custom.denyNonAllowedMachineTypes] :

student201998059@cloudshell:~ (student-00568)$ gcloud compute instance-groups unmanaged crea
te ci-ig \
  --zone=us-central1-a
Created [https://www.googleapis.com/compute/v1/projects/student-00568/zones/us-central1-a/in
stanceGroups/ci-ig].
NAME: ci-ig
LOCATION: us-central1-a
SCOPE: zone
NETWORK:
MANAGED:
INSTANCES: 0
student201998059@cloudshell:~ (student-00568)$ gcloud compute instance-groups unmanaged add-
instances ci-ig \

```